

- 1 The cubic equation $x^3 + bx^2 + d = 0$ has roots α, β, γ , where $\alpha = \beta$ and $d \neq 0$.

(a) Show that $4b^3 + 27d = 0$. [5]

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(b) Given that $2\alpha^2 + \gamma^2 = 3b$, find the values of b and d . [3]

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